LIFTING A BOUNDED ABOVE COMPLEX OF PROJECTIVE MODULES OVER A NILPOTENT QUOTIENT TO THE RING ITSELF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 19468-19481 (canonical tag 09AR) and the counted proof is the proof environment at source lines 19483-19495, which immediately follows it and is its own proof.

Editorial note. The proof refers to the five properties of the abstract lifting statement by number rather than by content, and each numbered reference is a single sentence whose input is named explicitly; no step of the verification is left out, so the card is not an instance of an omitted argument.

Editorial note. The four references are two to results of this chapter, namely the lifting criterion used as the framework and the boundedness statement, and two to results of other chapters on Nakayama's lemma and on lifting projective modules; the kinds recorded are the type words appearing in the labels themselves.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble; the counted text does not use the appended environments, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- The lifting statement for a class of modules closed under the relevant operations applies once five properties hold: closure under projectivity, closure under summands, surjectivity modulo the ideal, lifting of projectives from the quotient and boundedness of the object. invoked at proof lines 19483-19488. Supplies the criterion that is then verified property by property for the class of all projective modules.
- Nakayama's lemma gives that a map of projective modules which is surjective modulo the ideal is surjective. invoked at proof lines 19486-19487. Establishes one of the five properties for the class of projectives.
- Every projective module over the quotient lifts to a projective module over the ring, and boundedness of an object can be checked modulo a nilpotent ideal. used at proof lines 19488-19494. Establishes the remaining two properties and hence the conclusion.

Source reference. Original lemma 0BCA: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19390–19390. The reference is the criterion applied at proof

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lines 19483-19485; the verification of its five properties, which is the content of this proof, is inside the counted span.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Nakayama's lemma is invoked at proof lines 19486-19487 to verify that a map of projectives surjective modulo the ideal is surjective; it is one of the five property checks inside the counted span.

Source reference. Original lemma 07LV: sources/stacks-chow/tags.txt, source lines 9705–9705. The reference supplies the lifting of projective modules over the quotient, used at proof lines 19488-19490 to verify the fourth property; the use made of it is inside the counted span.

Source reference. Original lemma 0H75: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17307–17307. The reference is applied at proof lines 19491-19494 to see that the object is bounded below, which is the last of the five properties; the conclusion drawn from it is inside the counted span.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let $E^\bullet$ be a complex of R/I-modules. Let K be an object of $D(R)$. Assume that*

1. *$E^\bullet$ is a bounded above complex of projective R/I-modules,*
2. *$K \otimes^{\mathbf{L}}_R R/I$ is represented by $E^\bullet$ in $D(R/I)$, and*
3. *I is a nilpotent ideal.*

Then there exists a bounded above complex $P^\bullet$ of projective R-modules representing K in $D(R)$ such that $P^\bullet \otimes_R R/I$ is isomorphic to $E^\bullet$.

Proof. We apply Lemma 0BCA using the class $\mathcal{P}$ of all projective R-modules. Properties (1) and (2) of the lemma are immediate. Property (3) follows from Nakayama's lemma (Algebra, Lemma 00DV). Property (4) follows from the fact that we can lift projective R/I-modules to projective R-modules, see Algebra, Lemma 07LV. To see that (5) holds it suffices to show that K is in $D^-(R)$. Since we are given that $K \otimes^{\mathbf{L}}_R R/I$ is in $D^-(R/I)$ because $E^\bullet$ is bounded above, this follows from Lemma 0H75. □